

Ratio and Proportion

Ratio and Proportion. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

A ratio compares quantities — like the ingredients in a recipe. When you change the number of people, every ingredient has to change by the **same multiplier** to keep the taste right. This lesson helps you scale recipes, spot equivalent ratios, and share things fairly. The question you are exploring is: can maths help us share, scale and sustain things fairly?

What you are learning to notice

- How to read and write a ratio.
- That to scale, you multiply every part by the same number — not add the same amount.
- How to tell when two ratios are equivalent (the same relationship written differently).
- How to share an amount fairly in a given ratio.

Moving through it, one step at a time

Use the recipe scaler to predict amounts before you check them. Notice the multiplier that takes you from one column of a ratio table to the next, and that it acts on every ingredient. When you share an amount, add the parts of the ratio first to see how many equal shares there are. There is no rush — try a prediction, test it, and revise.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (scaling by a multiplier, equivalent fractions, and sharing into parts), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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